

Generative AI: An essential Skill for today's Data Analysts

Introduction

As a beginner in data analytics, you're stepping into a field that's rapidly evolving. Generative AI is becoming an essential tool for data analysts, allowing them to create new content and gain deeper insights. Let's explore what generative AI is and how it can enhance your skills.

What is generative AI?

Generative AI refers to a class of artificial intelligence models that create new content such as text, images, music, and more by learning patterns from existing data.

Generative AI can respond naturally to human conversation and serve as a tool for customer service and personalization of customer workflows. For example, you can use AI-powered chatbots, voice bots, and virtual assistants that respond more accurately to customers for first-contact resolution.

How does generative AI work?

Generative AI starts with a prompt that could be in the form of a text, an image, a video, a design, musical notes, or any input that the AI system can process. Various AI algorithms then return new content in response to the prompt. Content can include essays, solutions to problems, or [realistic fakes](#) created from pictures or audio of a person.

Early versions of generative AI required submitting data via an API or an otherwise complicated process. Developers had to familiarize themselves with special tools and write applications using languages such as Python.

Now, pioneers in generative AI are developing better user experiences that let you describe a request in plain language. After an initial response, you can also customize the results with feedback about the style, tone, and other elements you want the generated content to reflect.

Key techniques in generative AI:

Generative adversarial networks (GANs): GANs consist of two neural networks: the generator and the discriminator. The generator creates new data, whereas the discriminator evaluates it. Over time, the generator improves to produce realistic data.

Variational autoencoders (VAEs): VAEs encode input data into a compressed format and then decode it back, generating new data points similar to the input data.

Transformers: Used primarily in natural language processing (NLP), transformers generate human-like text by predicting the next word in a sequence. Generative Pre-trained Transformer 3 (GPT-3) is a notable example.

Generative AI models

Generative AI models combine various AI algorithms to represent and process content. For example, to generate text, various NLP techniques transform raw characters (e.g., letters, punctuation, and words) into sentences, parts of speech, entities, and actions, which are represented as vectors using multiple encoding techniques. Similarly, images are transformed into various visual elements, also expressed as vectors. One caution is that these techniques can also encode the biases, racism, deception, and puffery contained in the training data.

Once developers settle on a way to represent the world, they apply a particular neural network to generate new content in response to a query or prompt. Techniques such as GANs and VAEs—neural networks with a decoder and encoder—are suitable for generating realistic human faces, synthetic data for AI training, or even facsimiles of particular humans.

Recent progress in transformers, such as Google's Bidirectional Encoder Representations from Transformers ([BERT](#)), OpenAI's [GPT](#), and Google AlphaFold, have also resulted in neural networks that can not only encode language, images, and proteins but also generate new content.

What are the use cases for generative AI?

Generative AI can be applied in various use cases to generate virtually any kind of content. The technology is becoming more accessible to users of all kinds thanks to cutting-edge breakthroughs like GPT that can be tuned for different applications.

Some of the use cases for generative AI include the following:

- Implementing chatbots for customer service and technical support.
- Deploying deepfakes for mimicking people or even specific individuals.
- Improving dubbing for movies and educational content in different languages.
- Writing email responses, dating profiles, resumes, and term papers.
- Creating photorealistic art in a particular style.
- Improving product demonstration videos.
- Suggesting new drug compounds to test.
- [Designing physical products](#) and buildings.
- Optimizing new chip designs.
- Writing music in a specific style or tone.

What are the benefits of generative AI?

Generative AI can be applied extensively across many areas of the business. It can make it easier to interpret and understand existing content and automatically create new content. Developers are exploring ways that generative AI can improve existing workflows, with an eye to adapting workflows entirely to take advantage of the technology. Some of the potential benefits of implementing generative AI include the following:

- Automating the manual process of writing content.
- Reducing the effort of responding to emails.
- Improving the response to specific technical queries.
- Creating realistic representations of people.
- Summarizing complex information into a coherent narrative.
- Simplifying the process of creating content in a particular style

What are the limitations of generative AI?

Early implementations of generative AI vividly illustrate its many limitations. Some of the challenges generative AI presents result from the specific approaches used to implement particular use cases. For example, a summary of a complex topic is easier to read than an explanation that includes various sources supporting key points. The readability of the summary, however, comes at the expense of a user being able to vet where the information comes from.

Here are some of the limitations to consider when implementing or using a generative AI app:

- It does not always identify the source of content.
- It can be challenging to assess the bias of original sources.
- Realistic-sounding content makes it harder to identify inaccurate information.
- It can be difficult to understand how to tune in to new circumstances.
- Results can gloss over bias, prejudice, and hatred.

What are the concerns surrounding generative AI?

The rise of generative AI is also fueling various concerns. These relate to the quality of results, the potential for misuse and abuse, and the potential to disrupt existing business models. Here are some of the specific types of problematic issues posed by the current state of generative AI:

- It can provide inaccurate and misleading information.
- It is more difficult to trust without knowing the source and provenance of information.
- It can promote new kinds of plagiarism that ignore the rights of content creators and artists of original content.
- It might disrupt existing business models built around search engine optimization and advertising.
- It makes it easier to generate fake news.
- It makes it easier to claim that real photographic evidence of wrongdoing was just an AI-generated fake.
- It could impersonate people for more effective social engineering cyberattacks.
- Given the newness of GenAI tools and their rapid adoption, enterprises should prepare for the inevitable “trough of disillusionment” that’s part and parcel of emerging technology by adopting sound AI engineering practices and making responsible AI a cornerstone of their GenAI efforts, ensuring transparency, ethical considerations, and long-term sustainability in their AI implementations.

What are some examples of generative AI tools?

Generative AI tools exist for various modalities, such as text, imagery, music, code, and voices. Some popular AI content generators to explore include the following:

- Text generation tools include GPT, Jasper, AI-Writer, and Lex.
- Image generation tools include Dall-E 2, Midjourney, and Stable Diffusion.
- Music generation tools include Amper, Dadabots, and MuseNet.

- Code generation tools include codeStarter, Codex, [GitHub Copilot](#)[↗], and Tabnine.
- Voice synthesis tools include Descript, Listnr, and Podcast.ai.
- AI chip design [tool](#)[↗] companies include Synopsys, Cadence, Google, and NVIDIA.

Applications of generative AI in data analytics

Generative AI has many applications that can enhance your data analytics work:

Data augmentation: Create synthetic data to augment existing data sets, which is especially useful when data is scarce or imbalanced. This can improve predictive model performance.

Anomaly Detection: Identify anomalies or outliers by understanding the distribution of normal data. This is valuable in fraud detection, network security, and quality control.

Text and image generation: Generate realistic text and images for marketing, content creation, and customer engagement, such as automatic product descriptions and marketing visuals.

Simulation and forecasting: Simulate scenarios and forecast future events by generating potential outcomes from historical data. This is crucial in financial planning, supply chain management, and strategic decision-making.

Conclusion

Generative AI is a transformative technology that can significantly enhance your capabilities as a data analyst. By mastering generative AI techniques, you can unlock new possibilities in data augmentation, anomaly detection, content creation, and forecasting. As you embark on this journey, remember to balance innovation with ethical responsibility, ensuring that AI is used positively.